IOT-Note

**Module 1: Introduction to Peripherals and Interfacing**

- **Interfaces: These are intermediate components that connect the main computer system to peripherals (like HD monitors and speakers). They exist as Intermediate Hardware (like Timers, DMAs, and USB peripheral controllers) and Intermediate Software (like Device Drivers in Linux).**
- **Evolution of Peripherals: To save power, newer systems integrate peripheral controllers directly into the main processor chip (e.g., Intel Centrino integrates wireless controllers, and Intel Atom integrates graphics and memory controllers).**
- **Audio & Video Standards:**
    - **Dolby Digital 5.1: Uses 5 normal frequency channels (20-20Khz) and 1 low-frequency subwoofer (20-120Hz) with a max bit rate of 560 bit/s.**
    - **Dolby HD: 7.1 channels, Max bit rate of 18MB/s.**
    - **HD Cinema: Uncompressed high-definition video takes massive storage (e.g., 1080p is 1.35TB per hour), so compression formats like MP4 and MKV (matryoshka) are used.**

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8085 Microprocessor: 8085 is an 8-bit CPU that operates at 3-6MHz. it features a single cycle cpu structure and provide 16-bit address bus capable of accessing 64KB of Memory.

Register Structure of 8085: 8085 has 6 general purpose 8-bit registers (B,C,D,E,H,L) which can be paired to perform 16-bit operations(BC,DE,HL). Special types of registers of 8085 includes The accumulator, stack pointer, Program counter(stores the address of the next instruction to be fetched) and flag registers(flag, carry, zero, sign, parity and aux carry).

Instruction set categories:

Data transfer operations: used for loading and moving the data from one register to another. For example: MOV, MVI, LD, ST

Arithmetic: Used for basic arithmetic operations. For example: ADD, SUB, INR, DCR.

Logical Operation: Used for performing Logical operations. For example: AND, OR, XOR.

BRANCH Operations: Used for branching and jumping. For example: JUMP, CALL, JNZ.

8085 Bus and BUS Structure:

Data bus: Carries the actual data between memory, cpu and i/o. This is 8-bit in 8085. D0-d7

Address bus: Carries the address of the registers. in 8085 this bus is 16-bit which can carry 64KB of memory. A0-A15

Control bus: Carries the control signals such as read, write and enable between cpu, memory and I/O.

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Assembly Program to multiply two numbers 2000 and 2001

```
LDA 2000
MOV B,A
LDA 2001
MOV C,A
MVI A,00
L: ADD B
   DCR C
   JNZ L
STA 2012
HLT
```

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USB: Universal serial bus is a tree structure bus using 4-wire cables which are power, ground and 2 signal pins, replacing legacy serial/parallel ports. USB 1.1 operates at 1.5MBPS or 12MBPS, whereas USB 2.0 operates at 480MBPS with 125micro second of frames.

Encoding and Bit-Stuffing: USB uses NRZI(Non return to zero inverted) encoding, in this type of encoding, the signal transition only occurs if the next bit is zero. To prevent long strings of 1's from causing data loss, USB uses bit-stuffing, where a 0 is inserted after every six consecutive number of 1's.

USB transfer types: USB has 4types of transfers/endpoints:

1. control transfer: control transfer sets-up and configure the devices. It uses guaranteed 10% of the bandwidth.

2. Interrupt transfer: Transfers the interrupt signals, while polls at intervals of 1-255ms. For example mice and keyboard interrupt signal.

3. Bulk transfer: Large amount of data transfers with no data loss guaranteed. given that the low bandwidth is the priority. For example: Printers, Storage devices.

4. Isochronous Transfers: Isochronous transfers are used for real time applications where the speed of the data matters. For example Video or Audio. It guarantees constant data rate but has no error recovery.

USB Host Controllers: There are two types of host controllers in USB:

1. Universal Host Controller: This type of host controllers schedules the periodic transfers first, which can take upto 90% of the bandwidth. For example: Interrupts and Isochronous transfers.

2. Open Host Controller: Reserves space for non-periodic transfers first, which can take 10% of the bandwidth. For example: Bulk transfers and control transfers. then schedules the periodic transfers.

Linux USB Driver Hierarchy: Linux communicates asynchronously via URB(USB request block). the hardware hierarchy is bound as: Device->Configuration->Interface->Endpoints.

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**Module 4: Display Interfacing & Graphics**

- **LED & Character Displays:** Displays can be interfaced directly with the MPU without a peripheral controller using decoders (like a 3-to-8 decoder) and latches.
- **Refresh Rates:** The display is continuously written to at a high speed. The human eye cannot differentiate the ON/OFF state if the refresh rate is >30Hz (typically 30-60 Frames/Sec), making the blinking appear as a solid image.

- **Vector vs. Raster Graphics:**
    - **Vector:** Images are represented by mathematical equations/geometrical primitives (Points, Lines, Bézier curves like TrueType Fonts).
    - **Raster:** Images are represented by a grid of stored pixel values (BMP, PNG).
- **GPU Pipeline:** The GPU accelerates graphics operations through a fixed pipeline: **Programmable Vertex Processor -> Assembly -> Rasterization -> Fragment Processor -> Frame Buffer**

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8255 Programmable peripheral interface: used for dedicated i/o interface

ports: it has 2 8-bits ports(a and b) and one 8 bit port(c) divided into two 4-bits halves(upper-c and lower-c)

operating mode:

BSR (Bit set-reset): Controls port c bits individually. heavily used in handshaking and interrupts.

mode-0: used for simple/basic i/o for ports a,b and c.

mode-1: strobed i/o for ports a and/or b. port c is used for handshaking only.

mode-2: strobed bidirectional bus for port a only. port c is used for handshaking.

8251 USART (Universal synchronous/asynchronous receiver transmitter):

converts data bits from parallel to serial and from serial to parallel.

PINS: RxD (Receive Data), Txd(Transmit Data), RTS(Request to send), CTS(Clear to send), DTR(Data Terminal ready), DSR(Data set Ready).

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Arduino: Arduino is an open source physical programmable circuit board.

Key components: Power pins(5v,3.3v,Ground), Analog pins(A0-A5), Digital Pins(0-13), PWM(Pins for analog like outputs), AREF, Main-IC (ATMega), Tx-Rx LEDs, Reset Button.

RaspBerry-Pi: Raspberry-Pi is a credit card size single board computer that runs on linux-os

Key component: CPU(ARM-11), GPU(Used for better image Processing), SDRam Memory, Ethernet port, GPIO-Pins, Display and an SD Card slot for Mass Storage.

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IOT: Iot is the Interconnectedness of Physical Devices(e.g. Appliances,Vehicles) Embedded with softwares and sensors to connect and Exchange data.

M2M vs IOT: Machine to machine uses point-to-point data transfer, Has limited scope and uses traditional protocols.

Whereas IOT uses Internet-Protocols, Open-api and has larger scope as it uses cloud technology.

4-Stage IOT Architecture: IOT Uses 4-Layers architecture.

1. Sensing-Layer: This is the Layer from where Data Gathering from the outside physical environment occurs. Sensors and Actuators works in this layer.
2. Networking-Layer: In this Layer, Data Transmission between Multiple Devices using some protocols and Gateways take place.
3. Data-Processing-Layer: in Data Processing Layer, The Collected Data from Sensors are processed and Various types of control decision are take place based on the data. The work of this layer is happens on the CPU
4. Application-Layer: In this Layer, The user interacts with the System.

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Sensors And Actuators:

Sensors: Sensors collect the physical data from the environment and converts them into quantifiable signals.

Sensor Types Based on the independency:

1. Active Sensor: Active Sensors can Independently sense the data from the environment. For Example: Radar, Laser Altimeter.

2. Passive Sensor: Passive Sensor cannot independently sense the data from the environment. For example: Temperature, Accelerometer etc.

Sensor Types Based on Data output:

1. Analog Sensor: This type of sensor sends continuous outputs of the data. For example: LDR Sensor.

2. Digital Sensor: Digital sensors send binary output of the sensing data. For example: PIR Sensor.

Sensor Types Based on measurement Data Type:

1. Scalar Sensor: This type of sensor can only sense the magnitude. For Example: Temperature, Turbidity Sensor.

2. Vector Sensor: This type of sensor can sense both the magnitude as well as the direction. For example: Accelerometer and Gyroscope Sensor

Actuators: Actuators are the Devices which translate the control signal into mechanical Force.

There are three types of Actuators:

1. Hydraulic: This type of actuator uses fluid to create force. It is used for high force and high speed machine. But is expensive. For example machine used in construction, Pascals machine.

2. Pneumatic: Pneumatic actuators works by the compression of air or vacuum. this are cost effective and durable. Example: Robotic Arms.

3. Electrical: Electrical actuators runs by electric energy. For example: DC Motor, Bulbs, LEDs etc.

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Data Transmission:

Serial vs Parallel:

Parallel is faster, but limited to short distances due to data skew, complexity and higher cost. Whereas Serial Transmission is cheaper, But slower in speed as it transmits only 1 bit at a time.

Synchronous: Sender and Receiver must sync before starting the transmission of the data. sender sends blocks of data. more effective but expensive.

Asynchronous: Character-oriented, Sender sends the start bit, then sends the actual 8-bit data and at last 1-2 stop bits. Asynchronous transmission doesn't requires sender and receive to synchronize.

Based on direction:

Simplex: Data transmission occurs only in a single direction always. For example cpu sends data to printer.

Half Duplex: Bi-directional data transfer occurs, but only in a single direction at a time. For example: walkie-talkie

Full duplex: Fully directional,means data can be transferred from both directions even at the same time. For example: Copying a file from a pen-drive while sending another file to the pen-drive.

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IOT-Gateways: act as a bridge between devices or sensors and the cloud.

Functions: Protocol translation, Data aggregation, local edge computing, local edge filtering and providing security.

Cloud computing: is essential for IOT, because it provides distant computing power, vast storage of data, higher encryption or security, low cost of ownership and seamless device-to-device communication.

Data management is crucial for tracking, monitoring and refining iot data. tools like AWS IOT device manager helps securely onboard, organize, monitor and send over the air(OTS) firmware updates to a fleet of iot devices.

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- **MQTT (Message Queuing Telemetry Transport):** Lightweight, machine-to-machine, publish-subscribe protocol ideal for low-bandwidth environments. Uses a broker/server. Packet format: Fixed header (2 bytes), Variable Header, Payload.
- **ZigBee:** IEEE 802.15.4 standard. Low-cost, low-power, low data rate (20-250 kbps), short-range (75-100m).
  - *Topologies:* Star, Mesh, Tree.
  - *Devices:* Coordinator, Router, End Device. *Example:* Home automation.
- **Bluetooth & BLE (Bluetooth Low Energy):** Short-range UHF radio technology creating Personal Area Networks (PANs). Operates in a master-slave relationship (piconet). BLE is highly optimized for low power consumption in battery-operated IoT devices, featuring sleep modes and rapid reconnections.
- **CoAP (Constrained Application Protocol):** Client-server protocol that acts as an HTTP equivalent for constrained IoT devices. It runs over UDP (asynchronous) and uses GET, POST, PUT, DELETE.
  - *Messages:* Confirmable (CON - needs ACK), Non-confirmable (NON), Acknowledgment (ACK), Reset.
- **UDP (User Datagram Protocol):** A simple, connectionless, unreliable transport layer protocol. Best for IoT applications needing swift, low-latency transfers of small packets over reliable delivery. It has a tiny 8-byte header (Source Port, Destination Port, Length, Checksum)

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Error Check.

1. Parity Check

a. Even Parity: when odd number of 1 make d7=1. (Send even number of 1)

b. Odd parity: when even number of 1 make d7=1. (Send odd number of 1)

2. CheckSum:

*Used for block of data.

*Sum all of the bytes without carry and their 2's Complement.

*Total Sum result should be zero

3. Cyclic redundancy code(CRC):

* Synchronize communication.

* stream of data can be represented by cyclic polynomial, that is divided by constant polynomial.

* reminder to set bits and send out as check for error.

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Interfaces types:

parallel interface: multi-line channel transmitting bits simultaneously. requires buses of data.

Serial: Streams of data bits are transmitted one at a time, simpler wiring, less crosscheck.

Communication types:

H2M: Human to machine, when human gives input to a machine. for example human controls an iot device. face recognition system, etc.

M2M: Machine to machine communication is when a machine interacts with another machine automatically without any human intervention. For example: Sensor is sensing the data to esp-32

M2H: Machine to human communication is occurred when the machine sends the data to human such as fire alarm or sensor output through a display.

H2h: When a human interacts with another human using tools. for example two humans  chatting with each other.